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Effects of source of supplementary trace minerals in pre- and postpartum diets on reproductive biology and performance in dairy cows

B. Mion,[1] G. Madureira,[1] J. F. W. Spricigo,[1]* K. King,[1] B. Van Winters,[1] J. LaMarre,[2] S. J. LeBlanc,[3] M. A. Steele,[1] and E. S. Ribeiro[1]†
[1]Department of Animal Biosciences, University of Guelph, Guelph, ON, N1G 2W1, Canada
[2]Department of Biomedical Sciences, University of Guelph, Guelph, ON, N1G 2W1, Canada
[3]Department of Population Medicine, University of Guelph, Guelph, ON, N1G 2W1, Canada

ABSTRACT

Our objectives were to evaluate the effects of complete replacement of inorganic salts of trace minerals (STM) with organic trace minerals (OTM) in both pre- and postpartum diets on ovarian dynamics, estrous behavior measured by sensors, preimplantation conceptus development, and reproductive performance in dairy cows. Pregnant cows and heifers (n = 273) were blocked by parity and body condition score and randomly assigned to either STM or OTM diets at 45 ± 3 d before their expected calving. Pre- and postpartum diets were formulated to meet 100% of recommended levels of each trace mineral in both treatments, taking into consideration both basal and supplemental levels. The final target concentrations of Co, Cu, Mn, Se, and Zn were, respectively, 0.25, 13.7, 40.0, 0.3, and 40.0 mg/kg in the prepartum diet, and 0.25, 15.7, 40.0, 0.3, and 63.0 mg/kg in the postpartum diet. The STM group was supplemented with Co, Cu, Mn, and Zn sulfates and sodium selenite, while the OTM group was supplemented with Co, Cu, Mn, and Zn proteinates and selenized yeast. Treatments continued until 156 d in milk (DIM) and were assigned to individual cows using automatic feeding gates. Starting at 21 DIM, ultrasonography examinations of the ovaries were performed weekly to determine the presence of a corpus luteum and postpartum resumption of ovarian cyclicity. Cows were presynchronized with 2 injections of $PGF_{2\alpha}$ at 42 and 56 DIM. Estrous behavior was monitored using electronic activity tags that indirectly measured walking activity. Cows detected in estrus after the second $PGF_{2\alpha}$ were inseminated, and those not detected in estrus by 67 DIM were enrolled in a synchronization program. Cows that returned to estrus after artificial insemination (AI) were reinseminated. Pregnancy diagnosis was performed 33 d after AI, and nonpregnant cows were resynchronized. Transcript expression of interferon-stimulated genes in peripheral blood leukocytes was performed in a subgroup of cows (STM, n = 67; OTM, n = 73) on d 19 after AI. A different subgroup of cows (28 STM, 29 OTM) received uterine flushing 15 d after AI for recovery of conceptuses and uterine fluid for analyses of transcriptomics and metabolomics, respectively. In addition, dominant follicle diameter, luteal size and blood flow, and concentration of progesterone in plasma were measured on d 0, 7, and 15 relative to AI. After flushing, $PGF_{2\alpha}$ was given and the dominant follicle was aspirated 2 d later to measure the concentration of trace minerals by mass spectrometry. Estrous behavior, size of the dominant follicle and corpus luteum, concentration of progesterone, time to pregnancy, and proportion of cows pregnant by 100 d of the breeding period did not differ between treatments. A greater proportion of cows supplemented with OTM had a corpus luteum detected before presynchronization (64.3 vs. 75.2%), and primiparous cows supplemented with OTM tended to resume cyclicity earlier than their STM counterparts. Cows supplemented with OTM had a greater concentration of Cu in follicular fluid than cows supplemented with STM (0.89 vs. 0.77 μg/mL, respectively). In pregnant multiparous cows, expression of receptor transporter protein 4 in peripheral blood leukocytes was 42% greater in the OTM group. Conceptuses of the 2 treatments had 589 differentially expressed transcripts, with many indicating advanced conceptus elongation and greater transcript expression of selenoproteins in the OTM group. In pregnant cows, 24 metabolites were more abundant in the uterine fluid of OTM, including spermidine, sucrose, and cholesterol. In conclusion, replacing STM with OTM caused modest improvements to resumption of ovarian cyclicity and important changes in preimplantation conceptus development, but it did not alter conception risk and pregnancy rate.

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*Current address: Departamento de Zootecnia, Escola de Veterinária e Zootecnia, Universidade Federal de Goiás, Goiania, GO, Brazil 74690-900.
†Corresponding author: eribeiro@uoguelph.ca

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INTRODUCTION

The transition period in dairy cows, generally defined as 3 wk before calving to 3 wk after calving (Grummer, 1995), imposes metabolic challenges to the animals that often reduce subsequent milk production and reproductive efficiencies. Energy, nutrients, and oxidative balances are often compromised during transition, which impairs immune cell function and increases susceptibility to diseases, further compromising nutrient intake and partition (Sordillo and Aitken, 2009; LeBlanc, 2020). Health problems and extensive mobilization of body reserves after parturition have enduring effects on reproductive performance, decreasing the subsequent pregnancy per AI and increasing the risk of pregnancy losses (Ospina et al., 2010; Carvalho et al., 2014; Ribeiro and Carvalho, 2017; Carvalho et al., 2019). Supplementation of trace minerals (**TM**) is a required practice in dairy cow nutrition and potentially benefits the oxidative balance and health of transition cows (Goff, 2018). Although TM represent a small part of the diet and are present in low concentration in tissues, they are required for several biological processes, including the activity of multiple enzymes and transcription factors (Goff, 2018). A supplementation strategy that effectively provides the necessary amounts of TM to tissues could reduce the negative impact of the metabolic challenges in the transition period and indirectly benefit lactation and reproductive performances.

In addition to indirect effects on reproduction, supplementation of TM can directly affect fertility by changing systemic TM status and the cellular biology of reproductive organs, gametes, and developing embryo and placenta (Hostetler et al., 2003). For instance, Se and selenoproteins, such as glutathione peroxidase (**GPX**)1, GPX3, selenoprotein S, and selenoprotein M, were detected in the follicular wall of small and large bovine follicles, and the expression of GPX1 in the granulosa cells and that of selenoprotein S in the granulosa and theca cells were found to be upregulated during the development of large healthy follicles (Ceko et al., 2015). In vitro studies demonstrated that Se modulates the proliferation of bovine granulosa cells and stimulates estradiol synthesis in cells harvested from small and large follicles (Basini and Tamanini, 2000). Moreover, supplementation of Se in gilts affected the metabolism of embryos, including the upregulation of genes important for global protein synthesis of preimplantation conceptuses (Dalto et al., 2015). Zinc and Cu also play important roles in female fertility. Dairy cows that established normal luteal activity by 45 DIM had greater concentrations of Zn and Cu in plasma than cows with abnormal luteal activity or delayed first ovulation after calving (Nazari et al., 2019). In addition, cows that became pregnant at the first breeding had greater concentrations of Zn and Cu in plasma than cows that failed to become pregnant and those that lost the pregnancy (Nazari et al., 2019). Zinc is critical for meiotic progression of mammalian oocytes and their activation after fertilization (Kim et al., 2011), and the supplementation of Zn in culture medium increased the inner cell mass and the total cell number of bovine embryos produced in vitro (Wooldridge et al., 2019). Moreover, Mn is a cofactor for the enzymatic conversions of mevalonic acid to squalene, which is a precursor for the synthesis of cholesterol and steroid hormones (Olson, 1965), and for the metalloenzyme arginase, which is involved in the synthesis of urea, polyamines, and AA (Ash, 2004). The concentrations of Zn, Cu, and Mn are several folds greater in conceptuses than in other reproductive tissues, which indicates that the conceptus preferentially accumulates these minerals and that they could be important for the development, growth, and survival of the conceptus (Hostetler et al., 2003).

A common source of supplementary TM is inorganic salts of TM (**STM**). However, the absorption of TM elements from STM is often compromised by antagonistic interactions in the digestive tract that reduce their bioavailability for gut absorption (Goff, 2018). Organic sources of TM (**OTM**; i.e., chelates between the mineral and an organic molecule) are an alternative to STM supplementation. Their structure reportedly protects the TM from most unwanted interactions in the digestive tract and improves the bioavailability of TM for absorption (Spears, 1996; Spears, 2003). The beneficial effects of adding OTM into the diets of dairy cows on shortening the interval from calving to first estrus, reduction of days open, and greater pregnancy per AI have been previously reported (Ballantine et al., 2002; Nocek et al., 2006; Thatcher et al., 2010), although the results are not consistent (Griffiths et al., 2007; Rutigliano et al., 2008; Siciliano-Jones et al., 2008; Formigoni et al., 2011). This lack of consistency might be due to the large variety of experimental designs and management conditions. In a meta-analysis performed by Rabiee et al. (2010), supplementation with OTM reduced the median days open and the number of services per conception and increased the hazard of pregnancy by 150 DIM; the interval from calving to first service and 21-d pregnancy rate were not affected. Nevertheless, the understanding of how different TM supplementation strategies can affect cow fertility is still limited.

Our objective was to evaluate the effects of complete replacement of STM by OTM in both pre- and postpartum diets on ovarian dynamics, estrous behavior, preimplantation conceptus development, and reproductive performance in dairy cows. We hypothesized that total replacement of STM by OTM improves systemic TM status in dairy cows, contributing to effective cellular processes in the ovaries and uterus and improving ovarian steroidogenesis, estrous behavior, conceptus development, and overall reproductive performance.

MATERIALS AND METHODS

This study is part of multiple companion studies by Mion et al. (2022, 2023b) and Ogilvie et al. (2022). All research methods were approved by the University of Guelph Animal Care Committee (Animal Utilization Protocol #4064).

Experimental Design and Treatments

A minimum sample size of 120 cows per treatment was calculated to allow for an 80% probability of detecting, at a 5% significance level, a 14-d difference in median time to pregnancy (43 vs. 29 d) in a follow-up period of 100 d after the end of the voluntary waiting period (56–156 DIM). The statistical power calculation was performed in the WinPepi program version 11.65 (Abramson, 2011).

The experiment was performed at the Ontario Dairy Research Centre (Elora, ON, Canada) between December 2018 and September 2020. Pregnant cows (n = 173) and pregnant heifers (n = 100) were blocked by parity and BCS and randomly assigned to either STM or OTM treatment. Cows assigned to STM (n = 136) were supplemented with Co, Cu, Mn, and Zn sulfates and sodium selenite. Cows assigned to OTM (n = 137) were supplemented with Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech). A nonselenized yeast product (Diamond V Yeast XP, Diamond V) was added to both treatments to avoid a cofounding factor by the presence of *Saccharomyces cerevisiae* in only one of the treatments. Cows were enrolled in small weekly groups at 45 ± 3 d before their expected calving date and remained in the assigned treatment until the end of the experiment at 156 DIM. In both treatment groups, cows received the same basal TMR during the prepartum period and the same basal TMR during the postpartum period, except for the source of supplementary TM. The target concentrations of TM in the prepartum diet were 0.25, 13.7, 40, 0.3, and 40 mg/kg for Co, Cu, Mn, Se, and Zn, respectively. The target concentrations of TM in the postpartum diet were 0.25, 15.7, 40, 0.3, and 63 mg/kg for Co, Cu, Mn, Se, and Zn, respectively. Basal contributions of feed ingredients were considered for supplementation of Cu, Mn, and Zn, but not for supplementation of Co and Se. A detailed description of pre- and postpartum diet compositions was reported in a companion article (Mion et al., 2022).

Cows were housed in freestall pens and the treatments were assigned by Insentec feed bins (Insentec B. V.). During the prepartum period, diets were mixed and offered once a day at 1030 h. Two days before the expected calving date or when signs of calving (swelling of the vulva and relaxing of pelvic ligaments) were apparent, cows were moved to individual maternity pens. For the individual stalls, diets were offered once a day at 1030 h, and both feed offered and refusals were weighed daily. Seven days after calving, lactating cows were moved to freestall pens equipped with Insentec feed bins. During the postpartum period, diets were provided twice a day, 60% at 0900 h and 40% at 1530 h. Throughout the experiment, the amount of feed offered to each cow was adjusted daily to allow for 8% of orts.

Reproductive Management

Cows were presynchronized with 2 injections of $PGF_{2\alpha}$ (250 μg/mL; Estrumate, Merck Animal Health) at 42 ± 3 and 56 ± 3 DIM. The end of the voluntary waiting period was defined as the day of the second injection of $PGF_{2\alpha}$. Cows were monitored for estrous behavior through visual observation by the farm staff and electronic activity tags fitted on collars placed around the cows' necks (Delpro, DeLaval). A total of 148 cows (STM, n = 73; OTM, n = 75) were detected in estrus and inseminated once at 0800 h. The remaining 109 cows (STM, n = 52; OTM, n = 57) not detected in estrus by 67 ± 3 DIM (11 d after the second $PGF_{2\alpha}$) received an injection of GnRH (50 μg/mL; Fertiline, Vetoquinol). Two additional injections of $PGF_{2\alpha}$ were administered 5 and 6 d after the GnRH injection. Two days after the last $PGF_{2\alpha}$, cows received an injection of GnRH and were timed AI. All inseminations were performed by the EastGen Artificial Insemination Service. Cows that returned to estrus after an AI were reinseminated at 0800 h. Pregnancy diagnosis was performed 33 ± 3 d after AI by ultrasonography examination of the uterus and detection of an embryo with a heartbeat (7.5 MHz, Easi Scan, BCF Technology, IMV Imaging). Pregnant cows were re-evaluated 60 ± 3 d after AI for reconfirmation of pregnancy. Cows diagnosed as nonpregnant at 33 or 60 ± 3 d after AI were resynchronized using the 5-d CIDR Cosynch program and an intravaginal device with progesterone (CIDR, Zoetis) as described by Bisinotto et al. (2010). Twenty-six cows left the herd before the first AI (STM, n = 16; OTM,

n = 10), and only the remaining 247 cows (STM, n = 120; OTM, n = 127) were included in the analysis of reproductive performance. Every cow that entered the reproductive program was followed for 100 d, and every cow diagnosed nonpregnant during this period was resynchronized.

Resumption of Ovarian Cyclicity

Resumption of ovarian cyclicity was assessed by transrectal ultrasonography (7.5 MHz, Easi Scan, BCF Technology) for examination of the ovaries, once a week on Tuesdays, starting at 21 ± 3 DIM. The day when a mature corpus luteum (**CL**) was observed for the first time was considered the day that a cow resumed ovarian cyclicity. Anovulation was defined as the absence of a mature CL in the most current and all previous examinations. The last examination was performed at 63 ± 3 DIM, before the first GnRH injection of the synchronization program. Fourteen cows died or were sold in the first month of lactation (STM, n = 10; OTM, n = 4), and only the remaining 259 cows (STM, n = 126; OTM, n = 133) were in the analysis of resumption of ovarian cyclicity.

Characteristics of Estrous Behavior

For the purposes of the study, characterization of estrous behavior [i.e., duration of estrus, peak of activity during estrus, and area under the curve (**AUC**) for the relative increase in activity] was performed for estrus events detected from 56 to 67 ± 3 DIM, between the second $PGF_{2\alpha}$ of the presynchronization protocol and the first GnRH injection of the synchronization program, when flagged by the activity monitoring system (Delpro, DeLaval). The electronic activity tags measured physical impulses from changes in acceleration caused by movements of the head and neck as count per hour (Løvendahl and Chagunda, 2010). The raw hourly counts were extracted from the activity monitoring system into an Excel file, and a baseline activity for each cow was calculated as the average of the hourly counts measured during the 7 d preceding the second $PGF_{2\alpha}$ injection. For the observation period, 6-h moving averages were used to smooth the variability of raw activity values and to resemble the calculations performed by the activity monitoring system to flag cows in heat. The relative increase in activity was calculated by subtracting the 6-h moving average from the baseline, then dividing the difference by the baseline and multiplying the result by 100. The beginning of an estrus event was defined as a relative increase in activity equal to or greater than 100% for at least 3 consecutive hours. The end of an estrus event was defined as a relative increase in activity smaller than 100% for at least 5 consecutive hours. The duration of the event was defined as the difference in hours between the beginning and the end of the estrus event. Peak activity was defined as the highest relative increase through the duration of the estrus event. The AUC was calculated by the trapezoid method (Purves, 1992) for the relative activity values during the estrus event. In total, 154 cows (STM, n = 76; OTM, n = 78) had an estrus detected in this period; however, 11 cows (STM, n = 6; OTM, n = 5) were excluded from the analysis because the activity monitoring system did not reach the threshold that characterized the onset of estrus in our calculations.

Peripheral Blood Leukocyte Isolation, RNA Extraction, and Transcript Expression

Peripheral blood leukocytes (**PBL**) were isolated 19 d after first or second AI in a subgroup of cows (STM, n = 67; OTM, n = 73) for investigation of the expression of interferon-stimulated genes (**ISG**) according to Ribeiro et al. (2014). Briefly, blood samples were collected in tubes containing EDTA (BD Vacutainer), immediately placed at 4°C, and centrifuged for 15 min at 1,500 × *g* at 4°C within 2 h of blood collection. The buffy coat fraction was collected and placed in a 15-mL conical tube (Thermo Fisher Scientific). Red cell lysis buffer (150 m*M* NH_4Cl, 10 m*M* $NaHCO_3$, and 1 m*M* EDTA; pH 7) was added to a final volume of 15 mL. After homogenization, tubes were incubated at room temperature for 5 min and then centrifuged for 10 min at 300 × *g* at 4°C. The supernatant was discarded, and the pellet was resuspended in 10 mL of the red cell lysis buffer and incubated for 5 min. Tubes were then centrifuged for 10 min at 300 × *g* at 4°C, the supernatant was discarded, and the pellet was washed with 5 mL of PBS. Tubes were again centrifuged for 10 min at 300 × *g* at 4°C, the supernatant was discarded, and the pellet of PBL was resuspended in 0.8 mL of Trizol (Life Technologies), transferred to a 2-mL tube, and stored at −80°C.

Next, RNA was extracted from PBL samples using an extraction kit (PureLink RNA Mini Kit, Invitrogen) according to the manufacturer's instructions. Incubation with DNase (Invitrogen) for 15 min was added to the protocol according to the manufacturer's instructions. The RNA concentration of all samples was measured using a NanoDrop spectrophotometer (NanoDrop Technologies), and a subgroup of 12 samples was randomly chosen to confirm the RNA integrity and absence of gDNA contamination (Bioanalyzer, Agilent) after the extraction procedure. Complementary DNA was synthetized from 1,000 ng of RNA (High Capacity cDNA Reverse Transcription kit, Applied Biosystem)

Table 1. Primer sequences for genes investigated by quantitative reverse-transcription PCR in peripheral blood leukocytes

Target gene	Gene name	NCBI[1] sequence	Primer	Primer sequence
ISG15	Interferon-stimulated gene 15	NM_174366	Forward	5′-GGTATGAGCTGAAGCAGTT-3′
			Reverse	5′-ACCTCCCTGCTGTCAAGGT-3′
RTP4	Receptor transporter protein 4	BC105539	Forward	5′-CACATGTACCTGGAGAACCAGA-3′
			Reverse	5′-AGGTAGCTCTGAAACCTTCCTG-3′
ACTB	β-Actin	AY141970	Forward	5′-CTGGACTTCGAGCAGGAGAT-3′
			Reverse	5′-GGATGTCGACGTCACACTTC-3′
RPL19	Ribosomal protein L19	NM_001040516	Forward	5′-ATCGATCGCCACATGTATCA-3′
			Reverse	5′-GCGTGCTTCCTTGGTCTTAG-3′

[1]NCBI = National Center for Biotechnology Information.

and used for real-time quantitative reverse transcription PCR (**qRT-PCR**; QuantaStudio Real Time PCR system, Thermo Fisher Scientific Inc.). Samples were run in duplicate, and each 20-μL reaction mixture consisted of cDNA, forward and reverse primers (Table 1), SYBR green supermix (Bio-Rad), and nuclease-free water. The PCR parameters used were the following: 3 min at 98°C for polymerase activation followed by 40 cycles of a 2-step qRT-PCR (10 s at 98°C for denaturation, 30 s at 60°C for annealing and extension). The efficiency of primers ranged from 96.5 to 99.5%.

Two target genes, interferon-stimulated gene 15 (*ISG15*) and receptor transporter protein 4 (*RTP4*), and 2 reference genes, β-actin (*ACTB*) and ribosomal protein L19 (*RPL19*), were investigated (Table 1). For each sample, all genes were assayed in the same qRT-PCR plate. The threshold cycle (**CT**) for each gene for each sample was obtained from the qRT-PCR system and transferred to an Excel file. For each sample, the geometric mean of CT values obtained for the 2 reference genes was calculated and used for normalization of target CT values. The normalized CT value or delta-CT value was obtained by subtracting the geometric mean of reference genes from the CT value of the target gene. Delta-CT values were then used for statistics. The relative expression values were obtained by raising 2 to the power of the delta-delta CT, which was obtained from the delta-CT least squares means difference in the statistics output. Nonpregnant cows supplemented with STM were used as the reference group for comparison. Confidence intervals were generated by raising 2 to the power of the lower and upper 95% confidence limits obtained from the delta-CT least squares means difference.

Ovarian Structures, Progesterone, and Uterine Flushing

The last 57 cows enrolled in the experiment (STM, n = 28 and OTM, n = 29) were subjected to uterine flushing on d 15 after AI for possible recovery of preimplantation conceptuses and uterine luminal fluid (**ULF**) samples. Cows were subjected to the same reproductive management described above, except that all of them received timed AI at the end of the synchronization program and were not inseminated after estrus detection. All timed AI preceding the uterine flushing were performed by the same researcher using semen of a single high fertility sire.

In the same subset of cows, ultrasonography of the ovaries was performed before AI to measure the diameter of the ovulatory follicle and on d 7 and 15 after AI to measure the diameter and blood flow by color Doppler of the resulting CL (Ibex Evo II, E. I. Medical Imaging). Luteal volume was calculated by the following formula: volume = $4/3 \times \pi \times R^3$, in which R represents the radius. When a CL cavity was present, the volume of the cavity was calculated using the same formula, and this volume was subtracted from the total CL volume. Luteal blood flow was classified by scores 0 to 4, whereby 0 = no vascularization; 1 = 5 to 25% of the luteal area vascularized; 2 = 30 to 40% of the luteal area vascularized; 3 = 45 to 50% of the luteal area vascularized; and 4 = >50% of the area vascularized (Pugliesi et al., 2017). Blood samples were collected into sterile heparinized tubes (BD Vacutainer) at AI and at 7 and 15 d after AI for analysis of progesterone concentration in plasma by radioimmunoassay (ImmuChemt Coated tube Progesterone, MP Biomedicals, ICN Pharmaceuticals Inc.) as described by King et al. (2021). All samples were run in a single assay. Standards were run in triplicate, and controls and samples were run in duplicate. Two controls (high and low progesterone) were distributed evenly 6 times over the assay. The average concentration and intraassay coefficient of variation were 0.54 ng/mL and 13.5% for the low progesterone control and 4.21 ng/mL and 4.0% for the high progesterone control. Five cows failed to have the estrous cycle synchronized (4 STM and 1 OTM) and were excluded from the analysis of ovarian structures and progesterone concentration.

Before the uterine flushing procedure, epidural anesthesia was performed using lidocaine 2%, and the vulva was thoroughly cleansed and sprayed with chlorhexi-

dine solution. The uterine horn ipsilateral to the CL was flushed by transcervical catheterization and an infusion of 50 mL of PBS solution (0.01 *M* phosphate buffer and 0.154 *M* sodium chloride, pH 7.4) with 0.1% (vol/vol) of polyvinyl alcohol according to Ribeiro et al. (2016b). The flushing solution was infused using a 60-mL syringe attached to the Foley catheter (Rusch, Teleflex Medical). The fluid was recovered using the same syringe, placed in a petri dish, and examined under a stereoscope to search for the presence of a conceptus. Up to 6 extra flushings were performed if a conceptus was not observed in previous flushings. The fluid recovered from the first flushing of each cow was centrifuged at 2,000 × *g* at 4°C for 15 min, and the supernatant was transferred to 5-mL tubes and stored at −80°C for posterior analyses of ULF composition. The recovered conceptuses were washed in fresh PBS, snap-frozen in liquid nitrogen, and stored at −80°C until RNA extraction.

Ten days after the uterine flushing, this subgroup of cows was then enrolled in the regular reproductive program and followed for 100 d to evaluate reproductive performance.

Transcriptomics of Conceptuses

The RNA was extracted from conceptus cells using the AllPrep DNA/RNA Micro kit (Qiagen) according to the manufacturer's instructions. The RNA concentration was measured by a fluorometer using a Qubit RNA HS Assay (ThermoFisher Scientific). The quality and concentration of RNA were assessed in all the samples (Agilent Bioanalyzer), and samples with RNA integrity lower than 7 or less than 300 ng of total RNA were excluded. All samples that passed the quality control (STM = 6; OTM = 8) were subjected to transcriptome analysis by The Centre for Applied Genomics of the Sick Kids Hospital (Toronto, ON, Canada). The RNA libraries were prepared by NEBNest Ultra II Directional RNA Library Prep Kit (Illumina Inc.). Briefly, 300 ng of total RNA was enriched for poly-A mRNA using magnetic oligo d(T)25 beads and fragmented into pieces of 200 to 300 bases for 10 min at 94°C. The RNA fragments were converted to double-stranded cDNA, end-repaired, and adenylated at 3′. For amplification of library fragments, initial denaturation was performed for 30 s at 98°C, followed by 12 cycles of 10 s at 98°C, 75 s at 65°C, and an extension step for 5 min at 65°C. Each sample was amplified with a different barcoded adapter to allow for multiple sequencing. The size of libraries was assessed by the Bioanalyzer 2100 DNA High Sensitivity chip (Agilent Technologies). The RNA libraries were quantified by qPCR using the Kapa Library Quantification Illumina/ABI Prism kit (Kapa Biosystems Inc., Roche). Libraries were pooled in equimolar quantities and paired-end sequenced in a single lane of a S4 flowcell (Illumina Inc.) using the NovaSeq 6000 sequencing platform (Illumina Inc.), targeting approximately 30 million paired-end reads of 150 bases in length per sample.

Composition of ULF in Pregnant Cows

The composition of complex lipids, oxylipins, and primary metabolites were evaluated in a subset of ULF samples collected on d 15 of pregnancy (STM, n = 6; OTM, n = 8). These ULF samples were the first uterine flushings performed in the same cows whose conceptuses were subjected to transcriptomics analysis. All analyses were performed at the West Coast Metabolomics Center at the University of California (Davis).

For analysis of complex lipids, ultra-high-pressure liquid chromatography quadrupole time of flight mass spectrometry (**UHPLC-QTOFMS**) was used. Lipids were extracted from ULF samples according to procedures described by Matyash et al. (2008). Briefly, methanol–methyl tertiary-butyl ether was added to 500 μL of ULF to separate the lipids from proteins and hydrophilic molecules. Tubes were vortexed for 20 s and centrifuged for 2 min at 14,000 × *g*. The upper phase was transferred to tubes, and the content was dried and reconstituted using 65 μL of methanol-toluene (9:1, vol/vol). The extract was transferred to 2 separate amber glass vials with micro-inserts and analyzed by UHPLC-QTOFMS. The column used was a Water Acquity UPLC CSH C18 (2.1 × 100 mm, 1.7 μm; Waters Corp.) and was held at 65°C at a flow rate of 0.6 mL/min. The mobile phase A was composed of 60:40 acetonitrile-water, 10 m*M* ammonium formiate, and 0.1% formic acid. The mobile phase B was composed of 90:10 (vol/vol) isopropanol-acetonitrile, 10 m*M* ammonium formiate, and 0.1% formic acid. For the positively charged lipids, 3 μL of the sample was injected at 4°C in the Agilent QTOF 6530a, using a resolution of R = 10,000. For the negatively charged lipids, 5 μL of the sample was injected at 4°C in the Agilent QTOF 6546, using a resolution of R = 20,000. The raw data were processed by the MassHunter Qual (Agilent) to find peaks in up to 300 chromatograms. The peak features were imported into the MassProfilerProfessional for the peak alignment to identify the peaks present in multiple chromatograms. If a peak was not present in at least 30% of the chromatograms, it was excluded from the analysis. Peaks were collated and constrained into a MassHunter quantification method using tandem MS information and the LipidBlast library to identify

the lipids with manual confirmation of adduct ions and spectral scoring accuracy. Data were normalized to the sum of the 109 internal standards representing the multiple classes of complex lipids and reported as normalized peak heights for statistical purposes.

The composition of oxylipins in ULF samples was evaluated by ultra performance liquid chromatography–tandem MS according to procedures described by Pedersen and Newman (2018). Briefly, samples were thawed on ice and vortexed, and 50 μL of the sample was then mixed with 25 μL of Steroid Surrogate standards (250 n*M* stock solution in methanol), 25 μL of internal standard solution [1,000 n*M*, 1-cyclohexyl ureido, 3-dodecanoic acid (**CUDA**):1-phenyl 3-hexadecanoic acid urea in 50:50 acetonitrile-methanol], 25 μL of antioxidant solution (0.2 mg/mL BHT-EDTA in 1:1 methanol-water), and 125 μL of acetonitrile-methanol (50:50). The internal standards used were 1-phenylurea-3-hexanoic acid, 6-keto prostaglandin F1α-d4, thromboxane B2-d4, prostaglandin F2α-d4, prostaglandin D2-d4, CUDA, leukotriene B4-d4, 10,11-dihydroxyheptadecanoid acid, 14,15-DiHETrE-d11, 20-HETE-d6, 9(S)-HODE-d4, 12(S)-HETE-d8, 5(S)-HETE-d8, and 12(13)-EpOME-d4. Samples were vortexed for 30 s and centrifuged at 6°C for 5 min at 15,000 × *g*. Approximately 240 μL of the supernatant was added to a filter (0.1 μm) and centrifuged for 5 min at 12,000 × *g*, and the samples were placed into 96-well plates. Liquid chromatography was performed using an Agilent 1290 Infinity liquid chromatographer system (Agilent Technologies). The injection volume was 5 μL using a 1.7-μm Acquity C18 BEH column, 150 mm × 2.1 mm, and a 0.2-μm stainless steel frit as a guard column (Waters Corp. Inc.). The mobile phase A was composed of 0.1% acetic acid, and mobile phase B was composed of 10:90 isopropanol-acetonitrile in 0.1% acetic acid. Analyses were performed at a flow rate of 0.25 mL/min and the column was held at 50°C. Analytes were ionized in a negative mode by electrospray ionization, and MS was performed using a API4000 quadrupole ion trap mass spectrometer (AB-SCIEX) for multi-reaction detection. The rate of internal standard and surrogate standard was used to determine surrogate recovery, and corrections were applied to the measured target analyte concentration. Oxylipins were further classified according to their precursor fatty acids, which included arachidonic acid, dihomo-γ-linoleic acid, eicosapentaenoic acid, linoleic acid, and oleic acid.

The composition of primary metabolites in ULF samples was evaluated by GC-MS, using the column Rtx-5Sil MS (0.25 mm × 30 m, 0.25 μm; Restek Corp.). The injection volume was 0.5 mL at a flow rate of 1 mL/min. The initial injection temperature was 50°C, which was increased by 12°C/s to 250°C. The initial oven temperature program was 50°C and held for 1 min, followed by an increase of 20°C/min to 330°C, and held constant for 3 min. Analyses were performed on a Leco Pegasus IV mass spectrophotometer. The mass resolution was 17 spectra/s from 80 to 500 Da at −70 eV ionization energy and 1,800 V detector voltage with a 230°C transfer line and a 250°C ion source. Raw data were preprocessed by ChromaTOF version 2.32. The data files were exported to a data server with absolute spectra intensities and further processed by a filtering algorithm implemented in the metabolomics BinBase database (http://binvestigate.fiehnlab.ucdavis.edu/). The BinBase algorithm used the following settings: validity of chromatogram (<10 peaks with intensity $>10^7$ counts/s), unbiased retention index marker detection (MS similarity > 800, the validity of intensity range for high m/z marker ions), and retention index calculation by fifth-order polynomial regression. The spectra were cut to 5% base peak abundance and matched to database entries from most to least abundant spectra using the following filters: retention index window of approximately 2,000 units (equivalent to approximately 2 s of retention time), validation of unique ions and apex masses (unique ion was included in apex masses and presented at >3% of base peak abundance), mass spectrum similarity fitted criteria dependent on peak purity and signal/noise ratios, and a final isomer filter. Data were normalized by the sum of all the peak heights for all identified metabolites.

Concentration of TM in the Follicular Fluid

At the end of the uterine flushing procedure, cows received an injection of $PGF_{2\alpha}$. Two days later, the dominant follicle was aspirated transvaginally to recover the follicular fluid for analysis of TM concentration. Epidural anesthesia was performed with lidocaine 2% before aspiration. The vulva was thoroughly cleansed and sprayed with chlorhexidine solution. Transvaginal aspiration was conducted using an 18G needle guided by ultrasonography (Ibex Evo II, E. I. Medical Imaging) using a 6- to 10-MHz convex transducer (MC8OPU, E. I. Medical Imaging). The aspiration needle was attached to a tubing system connected to a sterile syringe at the distal end that was used to establish negative pressure and to collect the follicular fluid. Follicular fluid samples were placed in ice immediately after collection and centrifuged at 1,500 × *g* within 2 h of collection. The supernatant was transfer to a 2-mL tube and frozen at −20°C until analysis. Trace mineral concentration in follicular fluid samples was performed at the Animal Health Laboratory, University of Guelph. Samples

were diluted with 1% HNO_3, 1% isopropanol, 0.01% TritonX-100, and 0.01% EDTA and quantified against a multi-element standard curve by inductively coupled plasma MS (Agilent 7900 ICP-MS). After follicle aspiration, cows received an injection of GnRH and were enrolled in a 100-d reproductive program for analyses of reproductive performance as described above.

Statistical Analyses

Binary data were evaluated by logistic regression using the GLIMMIX procedure of SAS on Demand (SAS Institute Inc.). Time to resume ovarian cyclicity and time to pregnancy were analyzed by Cox's proportional hazard model using the PHREG procedure of SAS. Survival plots were generated with MedCalc, v. 12.7.2 (MedCalc). Continuous data were evaluated by ANOVA using the GLIMMIX procedure of SAS. Statistical models included the fixed effects of treatment, parity, the interaction between treatment and parity, and period (1–5, each representing 3-mo interval of enrollment). For pregnancy per AI, the effects of AI technician, sire, and breeding type (after estrus detection or timed AI) were added to the statistical models. For ISG expression in PBL, the statistical models also included the effect of pregnancy status on d 33 after AI; the interaction between treatment and pregnancy status; the interaction between treatment, parity, and pregnancy status; and the effect of AI number (first or second AI). For repeated measures, the statistical model included fixed effects (day; interaction between day and treatment; interaction between day and parity; and interaction between day, treatment and parity) and the random effect of cow nested within treatment. The covariance structure (autoregressive 1, compound symmetry, Toeplitz, unstructured, and spatial power covariance) that resulted in the lowest Bayesian information criterion was selected. For every model, residuals were tested for normality and homogeneity of variance, and data were transformed as needed.

Quality control of transcriptome data was performed using FastQC v.0.11.5 (Babraham Bioinformatics). Adaptors were trimmed by Trim Galore v.0.5.0. (Babraham Bioinformatics) and reads shorter than 40 nucleotides were discarded. The quality of the trimmed reads was reassessed using FastQC v.0.11.5. Trimmed reads were screened for the presence of rRNA and mtRNA sequences using FastQ-Screen v.0.10.0 (Babraham Bioinformatics). Read distribution, positional read duplication, and strandedness of the alignments were assessed using the RSeQC package v. 2.6.2 (Wang et al., 2012). The reference genome used was the *Bos taurus* build ARS-UCD1.2 (INSDC Assembly GCA_002263795.2). Raw trimmed reads were aligned to the reference genome using STAR aligner v.2.6.0c (Dobin et al., 2013). The filtered STAR alignments were processed to extract raw read counts for genes using HTSeq-count v.0.6.1p2 (Anders et al., 2015). Reads that equally overlapped 2 genes were not counted for either gene to avoid introducing bias in the expression results. Initial minimal filtering of 10 read counts per gene in at least 4 samples was applied to the data set. Box plots were generated to visualize the distribution of the gene counts for all samples to identify sample or gene outliers. Exploratory analysis and differential gene expression analysis was performed using DESeq2 v. 1.26.0s. package on R v.3.6.1 (https://www.r-project.org). The statistical model included the effects of treatment and parity. Differentially expressed genes (**DEG**) were considered when the adjusted P-value was < 0.05 and fold change ≥ 1.5. Network, functional, and causal analyses were performed using the Ingenuity Pathway Analysis software (Qiagen). Emphasis was given to canonical pathways, predicted upstream regulators, and downstream effects with z-score < -2 or >2 and overlap P-value ≤ 0.05. In addition, the final list of DEG was compared with the list of transcripts shown to be associated with elongation of preimplantation conceptus reported by Ribeiro et al. (2016b).

Statistical analyses of peak heights of complex lipids and primary metabolites in ULF were performed separately using the MetaboAnalyst 5.0 (https://www.metaboanalyst.ca/). In both data sets, features with missing data in 50% or more of samples were excluded from analyses. The remaining missing values were replaced by a hypothetical number equivalent to one-fifth of the minimum positive value of each variable. Filtering of the data was performed based on the interquartile range to remove variables unlikely to be used when modeling the data. Data were normalized by the sum method, log10 transformed, and scaled by the auto-scaling method. Statistical differences between treatment were determined by ANOVA, and the parity was included as a covariate in the model. Concentration of individual oxylipins, classes of oxylipids, classes of complex lipids, and total lipids in ULF were performed by ANOVA using the GLIMMIX procedure of SAS on Demand (SAS Institute Inc.). The statistical models included the effect of treatment and parity. For analyses of ULF composition and conceptus transcriptome, the interaction between treatment and parity was not included in the statistical models because of insufficient statistical power to evaluate the effects of treatment within parity.

Table 2. Estrous cyclicity and pregnancy outcomes in cows supplemented with inorganic salts of trace minerals (STM) or organic trace minerals (OTM) in both pre- and postpartum diets

Item	Treatment[1]		AOR (CI)[2]	*P*-value
	STM	OTM		
Left before cyclicity outcomes, % (n/n)	7.4 (10/136)	2.9 (4/137)	0.34 (0.09–1.34)	0.12
Cyclic before presynchronization, % (n/n)	64.3 (81/126)	75.2 (100/133)	1.86 (1.05–3.30)	0.03
Cyclic before synchronization, % (n/n)	94.4 (119/126)	95.5 (127/133)	1.20 (0.39–3.75)	0.75
AI after estrus detection, % (n/n)	58.4 (73/125)	56.8 (75/132)	0.94 (0.56–1.58)	0.81
Left before breeding outcomes, % (n/n)	11.8 (16/136)	7.3 (10/137)	0.70 (0.27–1.82)	0.46
Pregnant, % (n/n)				
At first AI	29.2 (35/120)	33.1 (42/127)	1.02 (0.55–1.87)	0.95
By 100 d of breeding	70.8 (85/120)	69.3 (88/127)	0.85 (0.45–1.58)	0.59
Pregnancy losses, % (n/n)				
By d 60 of gestation	9.9 (8/81)	7.1 (6/85)	0.58 (0.15–2.21)	0.42
By end of gestation	16.9 (13/77)	11.5 (9/78)	0.47 (0.13–1.63)	0.23

[1]Cows assigned to STM received supplementation with Co, Cu, Mn, and Zn sulfates and sodium selenite. Cows assigned to OTM received supplementation with Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech).
[2]AOR = adjusted odds ratio.

RESULTS

Resumption of Ovarian Cyclicity

The proportion of cows excluded from ovarian cyclicity analysis did not differ between treatments ($P = 0.12$). Compared with cows supplemented with STM, a greater proportion of cows supplemented with OTM ($P = 0.03$) were cyclic before the onset of presynchronization protocol at 42 DIM (Table 2). This difference, however, was no longer present at the onset of the synchronization program at 67 DIM (Table 2). In the survival analyses, primiparous cows supplemented with OTM tended ($P = 0.07$) to resume ovarian cyclicity earlier than primiparous cows supplemented with STM [adjusted odds ratio (AOR) = 1.49 (0.98–2.27)], but no difference was observed between multiparous cows [AOR = 1.01 (0.74–1.39)] (Figure 1A).

Reproductive Performance

The proportion of cows detected in estrus after the second $PGF_{2\alpha}$ injection of the presynchronization protocol and consequently inseminated after estrus detection did not differ between treatments (Table 2). The proportion of cows excluded from the reproductive performance analysis did not differ between treatments (Table 2). Pregnancy per AI for first AI, the proportion of cows pregnant after 100 d of breeding period (end

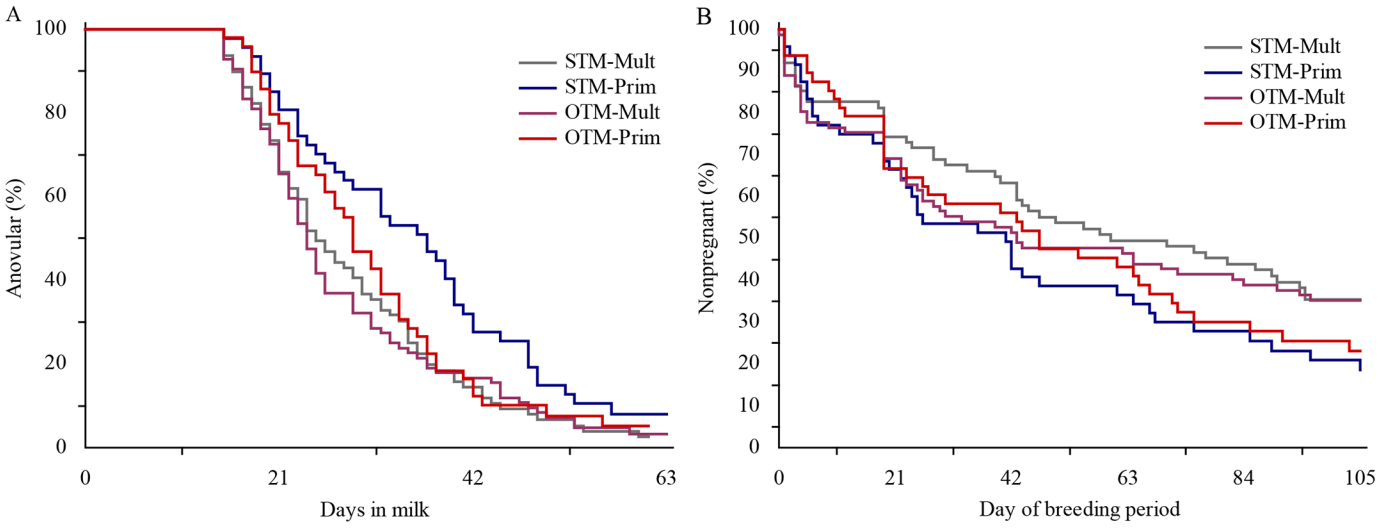


Figure 1. Survival probabilities for resumption to estrous cyclicity (A) and pregnancy (B) in primiparous (Prim) and multiparous (Mult) cows supplemented with inorganic salts of trace minerals (STM) or organic trace minerals (OTM) in both pre- and postpartum diets.

Table 3. Estrous behavior characteristics in cows supplemented with inorganic salts of trace minerals (STM) or organic trace minerals (OTM) in both pre- and postpartum diets

Item	Treatment[1]		*P*-value
	STM	OTM	
Number of estrus events evaluated	70	73	—
DIM on the day of estrus	58.41 ± 0.53	58.02 ± 0.52	0.59
Interval from $PGF_{2\alpha}$ injection to onset of estrus, h	89.43 ± 7.42	102.32 ± 7.21	0.21
Duration of estrus, h	11.78 ± 0.58	11.61 ± 0.57	0.84
Peak in relative activity during estrus	236.15 ± 11.60	236.84 ± 11.27	0.97
AUC[2] in relative activity during estrus	2,069.2 ± 156.8	2,061.51 ± 152.3	0.97

[1]Cows assigned to the inorganic group (STM) received supplementation of Co, Cu, Mn, and Zn sulfates and sodium selenite. Cows assigned to the organic group (OTM) received supplementation with Co, Cu, Mn, and Zn proteinates Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech).
[2]AUC = area under the curve.

of the study period), and pregnancy losses after d 33 of gestation did not differ between treatments (Table 2). We observed no effects of treatments or interaction between treatment and parity for time to pregnancy (Figure 1B).

Characteristics of Estrous Behavior

The proportion of cows detected in estrus by the software did not differ between treatments (STM = 60.9% vs. OTM = 59.5%). No differences regarding duration of estrus, peak in relative activity during estrus, and AUC of relative activity during estrus were observed between treatments (Table 3).

Transcript Expression of ISG in PBL

As expected, expression of the 2 ISG investigated in PBL samples on d 19 after AI were greater in cows diagnosed pregnant than in those diagnosed nonpregnant on d 33 after AI (Figure 2). More specifically, expression of *ISG15* was 4.7-fold greater ($P < 0.01$) and expression of *RTP4* was 2.2-fold greater ($P < 0.01$) in cows diagnosed pregnant than in cows diagnosed nonpregnant (Figure 2). For expression of *RTP4*, an interaction ($P = 0.03$) between treatment, parity, and pregnancy status on d 33 was observed (Figure 2A). For multiparous cows only, the increase in transcript expression in pregnant versus nonpregnant cows was greater ($P = 0.03$) in OTM (2.6-fold increase) than in STM (1.5-fold increase) (Figure 2A). However, this response was not observed in primiparous cows (Figure 2A). For expression of *ISG15*, no effects of treatment or interactions between treatment, parity, and pregnancy status were observed, although the pattern of expression was similar to that observed for *RTP4* (Figure 2B).

Ovarian Structures and Progesterone Concentration

No differences in size of the dominant follicle at AI, size and blood flow of CL on d 7 and 15 after AI, and concentration of progesterone in plasma on d 0, 7, and 15 relative to AI were observed between treatment groups (Table 4).

Transcriptome of Preimplantation Conceptuses

On average, 32.2 million reads per sample were generated in the RNA-seq analyses, and 79.2% of reads were uniquely mapped to the ARS_UCD1.2 bovine reference genome. Differential gene expression analysis revealed 589 DEG. Of these, 345 were upregulated and 254 were downregulated in conceptuses recovered from cows supplemented with OTM (Supplemental Table S1; https://doi.org/10.5281/zenodo.7839157; Mion et al., 2023a). The top 10 upregulated genes in the OTM group based on fold change were S100 calcium binding protein B (*S100B*), diacylglycerol kinase kapa (*DGKK*), phosphoglycerate mutase 2 (*PGAM2*), peroxisome proliferator-activate receptor gamma (*PPARG*), C-C motif chemokine ligant 17 (*CCL17*), pregnancy-associated glycoprotein 11 (*PAG11*), zinc finger protein 521 (*ZNF521*), Kruppel-like factor 13 (*KLF13*), myosin heavy chain 11 (*MYH11*), and gap junction protein β 6 (*GJB6*). The top 10 downregulated genes in the OTM group based on fold change were TBC1 domain family member 4 (*TBC1D4*), developmental pluripotency associated 2 (*DPPA2*), G protein subunit α 14 (*GNA14*), crumbs cell polarity complex component 2 (*CRB2*), ATPase phospholipid transporting 11A (*ATP11A*), inositol 1, 4, 5-trisphosphate receptor interacting domain containing 2 (*ITPRID2*), premature ovarian failure protein 1B (*POF1B*), cytoplasmic fragile X messenger ribonucleoprotein 1 interacting protein 2 (*CYFIP2*), ri-

Table 4. Characteristics of ovarian structures and concentration of progesterone in plasma in cows supplemented with inorganic salts of trace minerals (STM) or organic trace minerals (OTM) in both pre- and postpartum diets

	Treatment[1]		*P*-values of interest[2]		
Item	STM	OTM	TRT	TRT × Parity	TRT × Day
Diameter of dominant follicle at AI, mm	16.74 ± 1.06	18.68 ± 1.01	0.16	0.96	—
Volume of corpus luteum, cm^3			0.36	0.59	0.63
7 d after AI	6.95 ± 1.00	8.36 ± 0.92			
15 d after AI	7.68 ± 1.00	8.38 ± 0.92			
Blood flow of corpus luteum, scale 1–4[3]			0.32	0.65	0.63
7 d after AI	2.60 ± 0.17	2.68 ± 0.16			
15 d after AI	2.81 ± 0.17	3.04 ± 0.16			
Progesterone in plasma, ng/mL			0.43	0.08	0.54
At AI	0.14 ± 0.27	0.21 ± 0.23			
7 d after AI	2.42 ± 0.27	2.08 ± 0.23			
15 d after AI	4.96 ± 0.27	4.60 ± 0.23			

[1]Cows assigned to STM received supplementation with Co, Cu, Mn, and Zn sulfates and sodium selenite. Cows assigned to OTM received supplementation with Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech).

[2]Probability values for independent variables of interest: TRT = effect of treatment (STM vs. OTM); TRT × Parity = interaction between treatment and parity (multiparous or primiparous); TRT × Day = interaction between treatment and day of evaluation.

[3]Blood flow was measured as percentage of the luteal area that was vascularized using increments of 5 percentage points and then score as: 0 = <5% of the luteal area vascularized; 1 = 5 to 25% of the luteal area vascularized; 2 = 30 to 40% of the luteal area vascularized; 3 = 45 to 50% of the luteal area vascularized; and 4 = >50% of the area vascularized.

bosomal modification protein RimK-like family member B (*RIMKLB*), and acid phosphatase 3 (*ACP3*). Of the annotated DEG, 86 were also reported to be associated with the elongation of the preimplantation conceptuses in a study that used the same criteria for designation of DEG (Ribeiro et al., 2016b). Of these 86 genes, 50 that had previously been reported to have increased expression during elongation were upregulated in OTM conceptuses, and 33 genes reported to have reduced expression during elongation were downregulated in

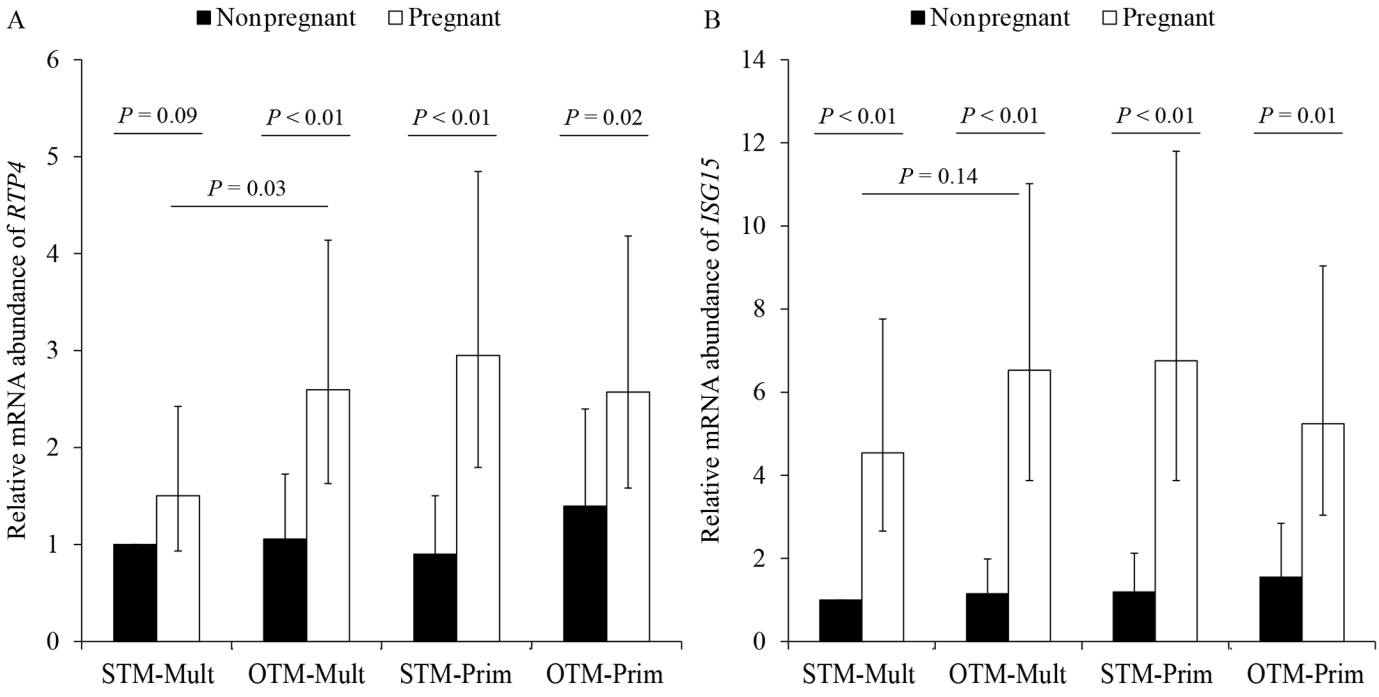


Figure 2. Relative mRNA expression of interferon-stimulated genes *RTP4* (A) and *ISG15* (B) in peripheral blood leukocytes collected on d 19 after AI from primiparous (Prim) and multiparous (Mult) cows supplemented with inorganic salts of trace minerals (STM) or organic trace minerals (OTM) in both pre- and postpartum diets according to their pregnancy status on d 33 after AI. Multiparous nonpregnant cows in the STM group were used as reference for comparison. Error bars represent the 95% confidence intervals for the group means.

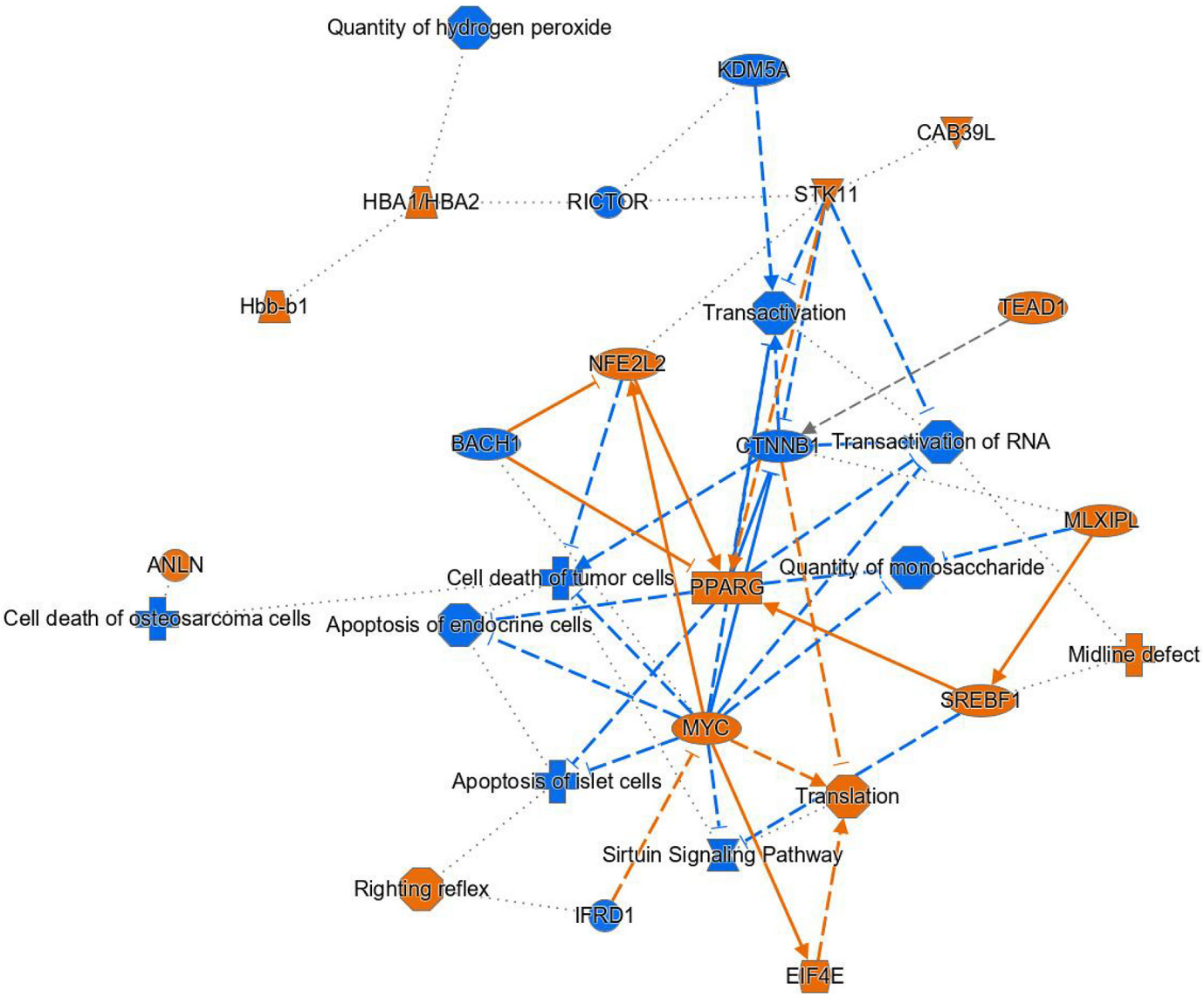


Figure 3. Unedited graphical summary of functional analysis retrieved from Ingenuity Pathway Analysis using information of 551 annotated differentially expressed genes (adjusted $P < 0.05$ and fold change >1.5) between preimplantation conceptuses collected from cows supplemented with inorganic salts of trace minerals (STM) or organic trace minerals (OTM) in both pre- and postpartum diets. The STM group was used as the reference for comparisons. Nodes in orange are predicted to be activated in OTM conceptuses, and nodes in blue are predicted to be inhibited in OTM conceptuses. The intensity of the color is based on the magnitude of the z-score in the analysis.

OTM conceptuses (Supplemental Table S2; https://doi.org/10.5281/zenodo.7839157; Mion et al., 2023a). The exceptions were 3 genes [tRNA-yW synthesizing protein 1 homolog (*TYW1*), retroelement silencing factor 1 (*RESF1*), and PR/SET domain (*2PRDM2*)] that were downregulated in OTM conceptuses but upregulated during elongation.

In the IPA Core Analysis of 551 annotated DEG (320 upregulated and 231 downregulated in OTM), a machine learning method was used to create a synopsis of the functional analyses that combined the most significant entities (e.g., canonical pathways, upstream regulators, diseases, and biological functions) and how they relate to each other into a graphical summary (Figure 3). Among the major biological themes highlighted in the unedited graphical summary, quantity of hydrogen peroxide, cell death, the sirtuin signaling pathway, transactivation, quantity of monosaccharide, and the transcription factors catenin β 1 (CTNNB1), BTB domain and CNC homolog 1 (BACH1), and lysine demethylase (KDM5A) were predicted to be inhibited in OTM conceptuses compared with STM conceptuses (Figure 3). In addition, translation of proteins and the transcription factors PPARγ, nuclear factor erythroid

2-related factor 2 (NFE2L2), proto-oncogene (MYC), sterol regulatory element-binding transcription factor 1 (SREBF1), carbohydrate-responsive element-binding protein (MLXIPL), and TEA domain transcription factor 1 (TEAD1) were predicted to be activated in OTM conceptuses compared with STM conceptuses (Figure 3).

Functional analysis revealed 7 canonical pathways with a z-score of <-2.0 or >2.0 and overlap P-value ≤ 0.05: oxidative phosphorylation (31 out of 111 molecules, z-score = 5.57, $P = 2.65 \times 10^{-25}$, Figure 4A), sirtuin signaling pathway (28 out of 292 molecules, z-score = −3.27, $P = 1.68 \times 10^{-10}$, Figure 4B), eukaryotic initiation factor 2 (EIF2) signaling (23 out of 224 molecules, z-score = 3.21, $P = 1.95 \times 10^{-9}$), human epidermal growth factor receptor 2 (HER-2) signaling (17 out of 227 molecules, z-score = −2.32, $P = 1.89 \times 10^{-5}$), coordinated lysosomal expression and regulation (CLEAR) signaling pathway (16 out of 285 molecules, z-score = −2.50, $P = 2.79 \times 10^{-4}$), coronavirus pathogenesis pathway (12 out of 203 molecules, z-score = −2.31, $P = 0.002$), and the adenosine monophosphate-activated protein kinase (AMPK) signaling (11 out of 244 molecules, z-score = −2.33, $P = 0.02$). A positive z-score indicates a predicted activation of the canonical pathway in the OTM group compared with the control group STM. A negative z-score indicates a predicted inhibition of the canonical pathway in OTM compared with STM.

Upstream regulator analysis revealed 45 upstream regulators with predicted activation and 26 upstream regulators with predicted inhibition in OTM compared with the control group STM, with absolute z-score ≥ 2 and overlap P-value < 0.05 (Supplemental Table S3; https://doi.org/10.5281/zenodo.7839157; Mion et al., 2023a). After the upstream regulators were ranked based on z-scores, the top 5 upstream regulators with predicted activation were TEAD1 (z-score = 4.5, overlap $P = 8.5 \times 10^{-13}$), MLXIPL (z-score = 4.24, overlap $P = 6.78 \times 10^{-11}$), serine/threonine kinase 11 (STK11, z-score = 4.11, overlap $P = 0.001$), retinoblastoma-associated protein (RB1, z-score = 3.84, overlap $P = 0.0001$), and calcium binding protein 39 like (CAB39L, z-score = 3.87, overlap $P = 4.0 \times 10^{-15}$). Likewise, the top 5 potential upstream regulators with predicted inhibition were rapamycin-insensitive companion of mammalian target of rapamycin (RICTOR, z-score = −6.85, overlap $P = 1.7 \times 10^{-25}$), La ribonucleoprotein 1 (LARP1, z-score = −4.24, overlap $P = 1.2 \times 10^{-12}$), lysine demethylase 5A (KDM5A, z-score = −4.24, overlap $P = 3.4 \times 10^{-9}$), carnitine palmitoyltransferase 1B (CPT1B, z-score = −4.12, overlap $P = 7.9 \times 10^{-11}$), and sirolimus (z-score = −4.03, overlap $P = 4.2 \times 10^{-6}$).

Regulator effect analysis revealed that the potential downstream effect of cell death of osteosarcoma cells and translation were regulated by 9 molecules [cluster of differentiation 3 (CD3), CTNNB1, eukaryotic translation initiation factor 4E (EIF4E), LARP1, LH, MLXIPL, MYC, MYCN, and RICTOR], which targeted 26 molecules in our data set [DEAD-box RNA helicase 3 (DDX3X), dipeptidyl peptidase 4 (DPP4), encoded ubiquitin-like protein (FAU), guanine nucleotide-binding protein (GNAS), Harvey murine sarcoma virus oncogene (HRAS), phosphatidylinositol 3-kinase (PIK3R1), PPARγ, proteasome 20S subunit α 7 (PSMA7), proteasome subunit β type-3 (PSMB3), putative RNA-binding protein 3 (RBM3), 60S ribosomal protein L11 (RPL11), ribosomal protein L27 (RPL27), ribosomal protein L30 (RPL30), ribosomal protein L35 (RPL35), ribosomal protein L37 (RPL37), ribosomal protein L38 (RPL38), ribosomal protein L39 (RPL39), ribosomal protein S15a (RPS15A), ribosomal protein S18 (RPS18), ribosomal protein S21 (RPS21), ribosomal protein S23 (RPS23), ribosomal protein S24 (RPS24), ribosomal protein S28 (RPS28), ribosomal protein S7 (RPS7), thioredoxin (TXN), and ubiquitin specific peptidase 18 (USP18)]. The familial congenital malformation was regulated by platelet-derived growth factor and vascular endothelial growth factors (PDGF BB), which targeted 9 molecules in our data set [aryl hydrocarbon receptor (AHR), calmodulin 1 (CALM1), cyclin D3 (CCND3), vascular endothelial growth factor receptor 1 (FLT1), FXYD, HRAS, lamin A/C (LMNA), MYH11, and PPARγ]. The quantity of monosaccharides was regulated by PPARγ, which targeted 6 molecules [1-acyl-sn-glycerol-3-phosphate acyltransferase (AGPAT2), cell death inducing DFFA like effect A (CIDEA), fatty acid binding protein 5 (FABP5), hormone-sensitive lipase E (LIPE), monoacylglycerol O-acyltransferase 1 (MOGAT1), and sirtuin 1 (SIRT1)], and by CPT1B, which targeted 5 molecules in the data set [AGPAT2, LIPE, pantothenate kinase 1 (PANK1), PIK3R1, and regulatory associated protein of MTOR complex 1 [RPTOR]). Transactivation was regulated by PDGF BB, which targeted 5 molecules [AHR, CCND3, HRAS, PPARγ, and proto-oncogene spi-1 (SPI1)], and by NFE2L2, which targeted 5 molecules in the data set [AHT, PPARγ, peroxiredoxin 1 (PRDX1), transcription factor A mitochondrial (TFAM), and TXN]. The potential downstream effect of transactivation of RNA was regulated by IL33, which targeted 6 molecules in the data set [ATP-binding cassette protein (ARB1), nuclear factor kappa β inhibitor β (NFKBIB), RNA polymerase I, II, and III subunit L (POLR2L), PPARγ, PRDX1, and SPI1].

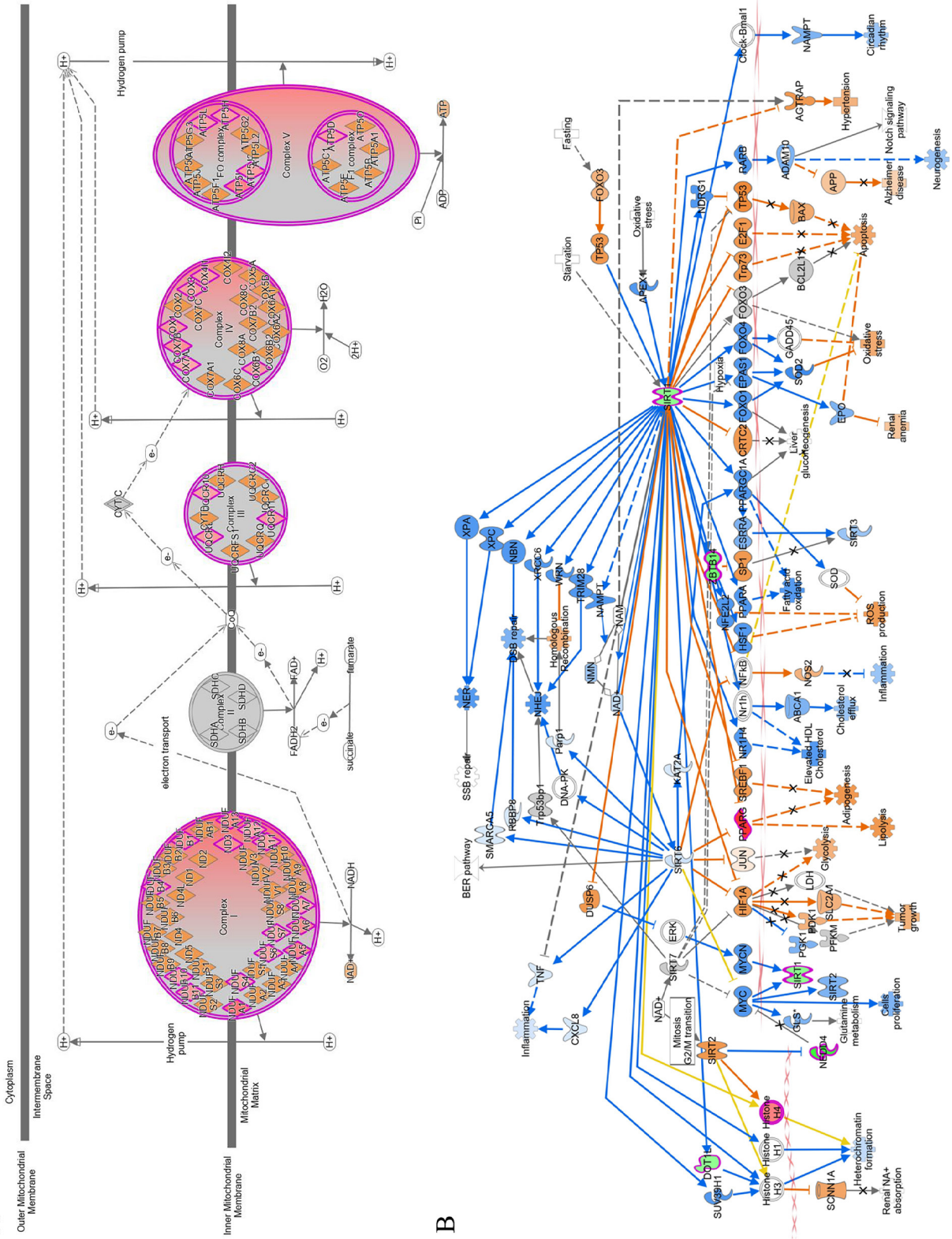


Figure 4. Top 2 canonical pathways associated with differences in transcriptome between preimplantation conceptuses collected from cows supplemented with inorganic salts of trace minerals (STM) or organic trace minerals (OTM) in both pre- and postpartum diets. (A) The oxidative phosphorylation pathway (31 out of 111 molecules, *z*-score: 5.6, $P = 2.65 \times 10^{-25}$). (B) The nuclear portion of the sirtuin signaling pathway (28 out of 292 molecules, *z*-score: −3.27, $P = 1.7 \times 10^{-10}$). Both figures were retrieved from Ingenuity Pathway Analysis, and the STM group was used as the reference for comparison. Color of nodes: green = downregulated; blue = predicted inhibition; pink = upregulated; orange = predicted activation.

Table 5. Complex lipids differentially abundant in the uterine luminal fluid collected on d 15 of pregnancy from cows supplemented with inorganic salts of trace minerals (STM) or organic trace minerals (OTM) in both pre- and postpartum diets[1]

Feature	ESI mode[2]	Mass to charge ratio, m/z	Retention time, min	Fold changes in OTM vs. STM	P-value
Sphingomyelin d40:3	Negative	841.65	5.78	2.62	0.01
Ceramide d34:0	Positive	540.54	5.83	2.34	0.03
Sphingomyelin 42:3	Positive	833.65	6.14	2.21	0.04
Physeteric acid, FA 14:1	Negative	225.19	1.18	2.18	0.04
Glucosylceramide d42:2	Negative	868.69	7.27	2.18	0.04
Cholesterol	Positive	369.35	4.83	2.13	0.05
Sphingomyelin 43:1	Negative	887.72	7.95	2.12	0.05

[1]Cows assigned to STM received supplementation with Co, Cu, Mn, and Zn sulfates and sodium selenite. Cows assigned to OTM received supplementation with Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech).
[2]ESI = electrospray ionization.

Complex Lipids in the ULF of Pregnant Cows

A total of 493 complex lipids were consistently identified in the ULF samples. Of these, only 7 lipids were affected by treatment, and all had increased concentration in pregnant cows supplemented with OTM in comparison with pregnant cows supplemented with STM (Table 5).

Oxylipids in the ULF of Pregnant Cows

Thirty oxylipids were consistently identified in the ULF samples. The concentration of individual oxylipins or groups of oxylipins and the percentage of individual oxylipins or groups of oxylipins did not differ between treatments (Supplemental Table S4; https://doi.org/10.5281/zenodo.7839157; Mion et al., 2023a).

Primary Metabolites in the ULF of Pregnant Cows

A total of 168 peaks were consistently identified in the ULF samples. Of these, 19 features were affected by treatment (Table 6). Two features had decreased concentration and 17 features had increased concentration in pregnant cows supplemented with OTM compared with pregnant cows supplemented with STM (Table 6). Pathway analysis revealed that differentially expressed metabolites were related to

Table 6. Primary metabolites differentially abundant in the uterine luminal fluid collected of d 15 of pregnancy from cows supplemented with inorganic salts of trace minerals (STM) or organic trace minerals (OTM) in both pre- and postpartum diets[1]

Feature	Mass to charge ratio, m/z	Retention index	Fold changes in OTM vs. STM	P-value
Nicotinamide	179	471,602	0.37	<0.01
Phenaceturic acid	91	654,351	0.41	0.02
Glycolic acid	177	227,636	2.12	0.05
Spermidine	144	792,924	2.13	0.05
2-Ketoadipic acid	100	331,841	2.19	0.04
Adipic acid	111	474,435	2.24	0.03
Terephthalic acid	295	611,213	2.30	0.03
1-Monopalmitin	371	900,644	2.32	0.03
Suberic acid	169	567,452	2.42	0.02
Isobutene glycol	129	229,353	2.43	0.02
Sucrose	271	912,704	2.58	0.01
β-Glycerolphosphate	243	574,470	2.59	0.01
Conduritol-β-epoxide	318	674,500	2.62	0.01
Galactinol	204	1,017,580	2.62	0.01
Raffinose	361	1,120,886	2.80	<0.01
4-Aminobutyric acid	304	488,730	2.84	<0.01
Glycerol-3-galactoside	204	800,205	2.88	<0.01
Glucoheptulose	217	828,606	2.94	<0.01
3-Aminoisobutyric acid	248	452,655	2.96	<0.01

[1]Cows assigned to STM received supplementation with Co, Cu, Mn, and Zn sulfates and sodium selenite. Cows assigned to OTM received supplementation with Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech).

Table 7. Concentration of trace minerals in follicular fluid collected from cows supplemented with inorganic salts of trace minerals (STM) or organic trace minerals (OTM) in both pre- and postpartum diets[1]

Item	Treatment		*P*-value
	STM	OTM	
Co, ng/mL	0.41 ± 0.05	0.42 ± 0.05	0.60
Cu, μg/mL	0.77 ± 0.04	0.89 ± 0.03	0.02
Mn, ng/mL	4.01 ± 0.60	2.80 ± 0.52	0.11
Se, μg/mL	0.08 ± 0.003	0.08 ± 0.003	0.47
Zn, μg/mL	0.85 ± 0.03	0.78 ± 0.03	0.09

[1]Cows assigned to STM received supplementation with Co, Cu, Mn, and Zn sulfates and sodium selenite. Cows assigned to OTM received supplementation with Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech).

galactose metabolism (3 out of 27 hits, $P < 0.01$) and arginine and proline metabolism (2 out of 38 hits, $P = 0.04$).

Concentration of TM in Ovarian Follicular Fluid

Cows supplemented with OTM had a greater concentration of Cu ($P = 0.02$) in follicular fluid than cows supplemented with STM (Table 7). However, concentrations of Co, Mn, Se, and Zn in follicular fluid did not differ between treatments (Table 7).

DISCUSSION

Trace minerals exert effects on multiple functions in the biology of mammalian cells, including immune cells (Spears and Weiss, 2008) and cells of reproductive tissues, gametes, and embryos (Kim et al., 2011; Ceko et al., 2015; Dalto et al., 2015; Wooldridge et al., 2019). Thus, different strategies of dietary supplementation of TM can potentially change the amount of TM available systemically or in specific tissues and cause biological effects. Here, we tested the hypothesis that complete replacement of STM by OTM would benefit systemic TM status, improving postpartum health and reproductive function of dairy cows. The latter could be a consequence of direct effects on reproductive tissues, the oocyte, and the embryo, and indirect effects related to enhancements of transition cow health. In a companion study, Mion et al. (2023b) reported a lower incidence of lameness and postpartum metabolic problems in addition to changes in the concentrations of Se and Co in multiple body fluids. In the current study, we observed that supplementation with OTM caused important changes in the biology of trophectoderm cells that were indicative of enhanced conceptus development, a modest increase in the concentration of Cu in follicular fluid, and earlier resumption of ovarian cyclicity, but no differences in time to pregnancy and conception risk.

The lack of differences in reproductive efficiency in the current study is supported by some previous studies (Griffiths et al., 2007; Rutigliano et al., 2008; Siciliano-Jones et al., 2008; Formigoni et al., 2011) but is in contrast with others that reported improvements in reproductive efficiency with the use of OTM supplements (Ballantine et al., 2002; Nocek et al., 2006; Rabiee et al., 2010; Thatcher et al., 2010). These inconsistent results might be related to differences in management conditions, diets formulations, level of milk production or oxidative stress of cows, or differences in supplementation strategies (i.e., top-up, partial, or complete replacement, level of supplementation, window of supplementation), the TM included in the experimental design, or the specific products and methods used. Complete replacement of STM (0.5, 21.4, 86, and 102 mg/kg of DM of Co, Cu, Mn, and Zn) by OTM in a concentrate pellet fed through a computerized feeder from 3 wk prepartum to 35 wk postpartum did not affect days to first service, number of days open, and services or conception in a study with 250 cows enrolled, although improvements in hoof health were observed (Siciliano-Jones et al., 2008). Similarly, the complete replacement of sodium selenite by selenized yeast (0.3 mg of Se/kg of DM) from 25 d prepartum to 80 DIM did not affect pregnancy rate at the first 2 AI in a study with 512 cows that were group-fed in 4 prepartum pens and 4 postpartum pens (Rutigliano et al., 2008). Partial replacement of STM (Cu, Mn, and Zn) by OTM from 60 d prepartum (16.9, 42.4, and 35.5 mg/kg of DM, respectively) to 240 DIM (10.4, 31.3, and 51.3 mg/kg of DM, respectively) resulted in similar time to first estrus, time to first service, proportion of cows pregnant, and days open in a study with 296 cows group-fed in 4 prepartum pens and 4 postpartum pens (Formigoni et al., 2011). In a study with grazing cows, Griffiths et al. (2007) divided 555 cows into 2 homogeneous groups that cycled through multiple pasture paddocks. The group assigned to OTM supplementation (12 mg of Co, 125 mg of Cu, 200 mg of Mn, and 360 mg of Zn) via concentrate or water from 35 d before calving to 230 DIM showed no differences in service rate, conception in the first AI, and number of nonpregnant cows compared with the other group of cows that received no supplementary TM, although a greater proportion of nonsupplemented cows received hormonal treatments for inactive ovaries at the onset of the breeding season. Nocek et al. (2006) used 4 prepartum pens and 8 postpartum pens to assign 573 cows into 4 supplementary TM strategies through 2 lactation cycles. Complete replacement of STM (Co, Cu, Mn, and Zn) by OTM,

feeding 100% of the NRC (2001) recommendations, resulted in no differences in days to first estrus, days to first service, services per conception, days open, and calving interval in lactation 1 and lactation 2. Partial replacement of Cu and Zn sulfates with Cu and Zn chelates associated with a higher inclusion of Co and Mn resulted in a shorter interval from calving to first estrus in both lactations, and a higher proportion of cows pregnant at first service and by 150 DIM in lactation 2 only. Ballantine et al. (2002) reported a reduction in days open (169 vs. 147) with complete replacement of STM (Co, Cu, Mn, Zn) by OTM (prepartum: 0.6, 26.7, 80.1, and 93.9 mg/kg of DM; postpartum: 1.6, 28.7, 127.5, and 169.4 mg/kg of DM, respectively) in a study with 300 cows that were group-fed from 21 d before calving to 250 DIM. A meta-analysis by Rabiee et al. (2010) reported that inclusion of OTM (Co, Cu, Mn, and Zn) in dairy cow diets reduced the median days open by 13.5 d and the number of services per conception by 0.27 and increased the relative risk of pregnancy at 150 DIM by 7%. No differences in time to first service and 21-d pregnancy rates were observed, and no evidence of heterogeneity was found for any of the reproductive variables evaluated. Finally, Thatcher et al. (2010) reported that complete replacement of sodium selenite by selenized yeast (0.3 mg of Se/kg of DM) resulted in greater pregnancy per AI in the second service in lactating cows under heat stress.

One of the possible mechanisms by which TM status or supplementation strategy could affect reproduction of dairy cows is through changes in ovarian biology. Trace minerals and metalloproteins are present in ovarian follicles and CL and are important for steroidogenesis (Olson, 1965; Basini and Tamanini, 2000; Ceko et al., 2015). Enhanced synthesis of estradiol by ovarian follicles could hasten the postpartum resumption of ovarian cyclicity, enhance the manifestation of estrous behavior, and benefit fertility of dairy cows, while greater synthesis of progesterone by the CL could improve oocyte quality and pregnancy development and survival. In addition to steroidogenesis, greater availability of TM in ovarian follicles could enhance the oxidative balance of follicular cells and oocytes (Picco et al., 2012) and improve the required activation of oocytes at fertilization (Kim et al., 2011). Nazari et al. (2019) reported that cows that established normal luteal activity by 45 DIM had greater concentrations of Zn and Cu in plasma than cows with abnormal luteal activity or delayed first ovulation after calving. In addition, cows that became pregnant at the first breeding postpartum had greater concentrations of Zn and Cu in plasma than cows that failed to become pregnant and those that lost pregnancy (Nazari et al., 2019). Moreover, studies have reported that supplementation of OTM shortens the interval from calving to first estrus (Campbell et al., 1999; Nocek et al., 2006), which indicates that cows supplemented with OTM could resume ovarian cyclicity earlier postpartum; however, this response is not consistent (Hackbart et al., 2010; Formigoni et al., 2011). Grazing cows supplemented with OTM had a smaller proportion of cows treated for small or inactive ovaries at the beginning of the breeding season than unsupplemented cows (Griffiths et al., 2007), and beef cows supplemented with OTM had a greater number of cumulus-oocyte complexes recovered from ovum pick-up sessions than beef cows supplemented with STM (Dantas et al., 2019). In our study, the concentration of Cu in follicular fluid was greater in cows supplemented with OTM, but no differences were observed in the concentrations of other TM. In addition, a greater proportion of OTM fed cows were cyclic by 42 DIM, before the onset of the $PGF_{2\alpha}$-based presynchronization program. This difference in cyclicity, however, was no longer present at 63 DIM, before the onset of the synchronization program. In the survival analyses, primiparous cows supplemented with OTM tended to resume ovarian cyclicity earlier than primiparous cows supplemented with STM, but no differences between treatments were observed in multiparous cows. It is noteworthy that primiparous cows supplemented with OTM had lower milk production and higher postpartum BCS than primiparous cows supplemented with STM (Mion et al. 2022), which might help to explain the differences in ovarian cyclicity described above, given the implications to energy partition (Butler, 2003; Santos et al., 2004). Moreover, no differences in sizes of ovarian structures, concentration of progesterone in plasma, and estrous behavior duration and intensity were observed. Thus, the effects TM treatments on ovarian biology were considered only modest.

In addition to ovarian biology, differences in TM status or supplementation strategies could potentially affect the biology of endometrium cells, the uterine environment, and consequently, the development of embryo and placenta. Using a conservative data analysis, we found major differences in the transcriptome of preimplantation conceptuses between the 2 dietary treatments. Comparison of DEG in the current study with the reported changes in gene expression of bovine preimplantation conceptuses during onset of elongation (Ribeiro et al., 2016b) revealed many genes in common. At least 86 DEG in the TM study were reported to undergo changes in expression patterns during the onset of elongation. Interestingly, 83 of these genes had an expression pattern indicating that OTM conceptuses were more advanced in development than STM conceptuses. Among those are genes that have transcript expression

increased during the onset of elongation such as *PPARG*, solute carrier family 10 member 1 (*SLC10A1*), STAR-related lipid transfer domain containing 10 (*STARD1*), *MOGAT1*, acyl-coA transferase 8 (*ACOT8*), *PAG11*, *PAG8*, and *S100B*; genes that were upregulated in OTM compared with STM conceptuses; and genes that have transcript expression decreased during the onset of elongation such as *SIRT1*, 2-aminoethanethiol dioxygenase (*ADO*), coagulation factor II receptor (*F2R*), lysophosphatidylcholine acyltransferase 1 (*LPCAT*), grancalcin (*GCA*), *TBC1D4*, ubiquitin protein ligase (*NEDD4*), and DNA-methyltransferase 3A (*DNMT3A*) that were downregulated in OTM compared with STM conceptuses. Some of these gene expression changes are thought to be essential for coordination of cellular biology changes needed for conceptus elongation, which is a prerequisite for maternal recognition of pregnancy, implantation, and survival of pregnancy (Ribeiro et al., 2016b,c; Spencer et al., 2016). For instance, PPARγ is a transcription factor that coordinates cellular differentiation of human trophectoderm cells (Fournier et al., 2011), elongation of ovine preimplantation conceptuses (Brooks et al., 2014), and lipid metabolism of ruminant conceptuses (Ribeiro et al., 2016c). It was listed as a possible upstream regulator of transcriptional changes observed in OTM conceptuses in this study. In endothelium cells, PPARγ activity has been reported to be modulated by Zn, with Zn deficiency reducing PPARγ activity (Meerarani et al., 2003; Reiterer et al., 2004). Moreover, a recent study in Se-deficient beef heifers reported large conceptus sizes on d 17 of gestation in heifers supplemented with a 50-50 mix of organic and inorganic Se compared with heifers supplemented 100% inorganic Se (Crites et al., 2022). Our prediction of more advanced development in OTM conceptuses is also supported by the observed higher expression of *RTP4* in PBL of OTM multiparous cows and changes in ULF composition. Higher expression of *RTP4* has been observed in pregnant cows that were healthy in the early postpartum period compared with pregnant cows that had postpartum clinical disease (Ribeiro et al., 2016a), and in pregnant cows supplemented with recombinant somatotropin during the preimplantation phase compared with pregnant cows in the unsupplemented control group (Ribeiro et al., 2014). In both models, the higher expression of ISG was followed by improvements in pregnancy and calving per AI.

Functional analyses of all annotated DEG suggested important differences in biology of OTM and STM conceptus cells. Conceptuses collected from cows supplemented with OTM were predicted to have reduced quantity of hydrogen peroxide and apoptosis of cells and greater translation of proteins compared with conceptuses collected from cows supplemented with STM. These predictions are likely related to the predicted activation of 2 canonical pathways: oxidative phosphorylation and EIF2 signaling. The former canonical pathway was predicted to be activated because of upregulation of 5 ATP synthase subunits, 8 cytochrome C oxidase subunits, 14 NADH-ubiquinoneoxidoreductase subunits, a NADH dehydrogenase subunit, 2 ubiquinol-cytochrome c reductases, and a ubiquinolcytochrome reductase binding protein. The latter canonical pathway was predicted to be active because of upregulation of 20 ribosomal proteins and 2 GTPases. Interestingly, a study that compared global gene expression of porcine blastocysts on d 5 of development collected from gilts supplemented with organic or inorganic sources of Se or an unsupplemented control group showed upregulation of genes involved with protein synthesis and antioxidant defense in the organic Se group compared with the inorganic Se and the control group (Dalto et al., 2015). Moreover, compared with the unsupplemented control group, supplementation with organic Se resulted in larger changes in the transcriptome of blastocysts than supplementation with inorganic Se (Dalto et al., 2015). Conversely, partial replacement of STM (Co, Cu, Mn, and Zn) by OTM from 40 to 72 d before calving through 69 to 116 d after calving did not alter the quality of bovine embryos collected on d 6 of development based on visual examination and assessment of accessory sperm and blastomere numbers (Hackbart et al., 2010). No molecular analyses of embryos were performed in the study by Hackbart et al. (2010).

Oxidative phosphorylation is crucial for the energy metabolism of cells, and the upregulation of 31 molecules acting in 4 out 5 complexes of the respiratory electron transport chain in OTM conceptuses likely increased synthesis of ATP and NAD^+, which are both required in large quantities by trophectoderm cells in the elongating conceptuses (Bazer et al., 2021). Transcriptome analysis of long noncoding RNA in ovine conceptuses collected on d 15, 17, 19, and 21 after estrus also predicted an upregulation of oxidative phosphorylation in trophectoderm cells of conceptuses that were in a more advanced stage of development (Matsuno et al., 2022). In addition, the study showed greater levels of NAD and NADH in more advanced conceptuses (Matsuno et al., 2022). Interestingly, pregnant cows supplemented with OTM had higher concentrations of multiple energy metabolites and lower concentration of nicotinamide in the ULF, which could be associated with the increased demand of energy substrates and use of nicotinamide for synthesis of NAD/NADH by the more advanced conceptuses. Moreover, the canonical pathway of sirtuin-1, a NAD-dependent deacetylase, was predicted to be inhibited in conceptuses of cows supplemented

with OTM. In fact, expression of *SIRT1* itself and several other genes reported to be regulated by sirtuin-1 were downregulated in OTM conceptuses. Sirtuin-1 has a central role connecting energy metabolism and cellular signaling and is considered an energy sensor in cells. Expression of sirtuin-1 increases during periods of caloric restriction and decreases in period of energy abundance (Finkel et al., 2009; Chang and Guarente, 2014). Ribeiro et al. (2016b,c) reported a reduction in expression of *SIRT1* in conceptus cells during onset of elongation, which was negatively correlated with expression of *PPARG*. In white adipose tissue, the negative regulation between SIRT1 and PPARγ in a feedback loop has been characterized (Picard et al., 2004; Han et al., 2010) and might also occur in trophectoderm cells of the elongating conceptus (Ribeiro et al., 2016b,c). Sirtuin-1 uses NAD^+ as a cosubstrate, which is cleaved into nicotinamide and ADP-ribose (Chang and Guarente, 2014). Thus, the inhibition of sirtuin-1 could also have contributed to the lower abundance of nicotinamide in the ULF of OTM cows.

During oxidative phosphorylation, a small portion of oxygen molecules are converted to superoxide anion radical in the complex I and III of the mitochondria, and the dismutation of superoxide anion generate hydrogen peroxide (Turrens, 1997). Interestingly, the quantity of hydrogen peroxide was predicted to be reduced in OTM conceptuses, mostly because of the upregulation of genes *PRDX1*, *PRDX5*, parkinsonism-associated deglycase (*PARK7*), NADH:ubiquinone oxidoreductase subunit A13 (*NDUFA13*), and cyclin-dependent kinase 5 (*CDK5*), whose proteins were all reported to be involved in reduction of hydrogen peroxide and oxidative stress. Nonetheless, molecules of hydrogen peroxide are also neutralized by enzymes of the GPX family, which have a molecule of Se in their active site. In the present study, we observed upregulation of genes that encode selenoproteins in conceptuses collected from cows supplemented with OTM, including *GPX4* and selenoprotein K (*SELENOK*). Greater GPX expression could help conceptuses avoid oxidative stress as a consequence of a highly functional mitochondria. Complete deletion of GPX4 in mouse embryos resulted in early embryonic lethality (Imai et al., 2003) likely because of the importance of GPX4 for proliferation and differentiation of developing tissues (Schneider et al., 2006). The upregulation of genes that encode selenoproteins could potentially be related to a greater concentration of Se reaching the uterine lumen in cows supplemented with OTM. In a companion study, Mion et al. (2023b) reported higher concentrations of Se in serum and milk of cows supplemented with OTM. Unfortunately, we were not able to accurately measure TM in ULF samples likely because of the dilution and possible interferences caused by the saline solution used for flushing of the uterus. However, preimplantation conceptuses are dependent on histotrophic nutrition from the endometrium, and cells of the conceptuses could also be seen as sensors of changes in the uterine environment, including differences in the availability of micronutrients.

Manganese can act as a divalent metal ion needed to stimulate the activity of enzymes involved in the synthesis of squalene (e.g., farnesyl pyrophosphate synthase, squalene synthase), which is a precursor for synthesis of sterols (Beytia et al., 1973; Liu et al., 2014; Luo et al., 2020). In addition, Mn is a cofactor of arginase, a metalloenzyme required for synthesis of polyamines, urea, and proline from arginine (Ash, 2004). Interestingly, we observed greater concentrations of cholesterol and spermidine in ULF samples collected from pregnant cows supplemented with OTM than in ULF samples collected from pregnant cows supplemented with STM. In addition to being a precursor for the synthesis of steroid hormones, bile acids, and vitamin D, cholesterol is essential for the structure and function of cell membranes (Mouritsen and Zuckermann, 2004) and is therefore needed for the rapid proliferation of trophectoderm cells and for the developing embryo proper in the elongating conceptus (Baardman et al., 2013; Ribeiro et al., 2016c). Polyamines such as spermidine are essential for cell signaling of the elongating conceptus, stimulating mTOR signaling, protein synthesis, cell proliferation and migration, and IFN-τ secretion (Halloran et al., 2021). Thus, the observed differences in concentration of cholesterol and spermidine in ULF in the current study could favor conceptus development in OTM cows and help explain the predicted advanced stage of development of OTM conceptuses. In addition to possible differences in activity of metalloenzymes, the reported differences in the metabolome of ULF could also be an indirect consequence of treatment effects in feed intake or energy balance.

Despite the large efforts to have 273 cows fed individually for a period of 6 to 7 mo, it is important to recognize some limitations of our study, including limited statistical power to evaluate reproductive performance, the fact that transcript expression of conceptuses and composition of ULF were evaluated only in a small number of samples, and the fact that our treatment design did not allow pinpointing the effects of individual TM, only the overall effects of the management strategy (i.e., the complete replacement of STM by OTM).

CONCLUSIONS

Complete replacement of supplementary STM by OTM in both pre- and postpartum diets of dairy cows

resulted in important differences in preimplantation conceptus development. These differences included major changes in transcript expression of trophectoderm cells that indicated alterations in energy metabolism and advanced conceptus development in cows supplemented with OTM, in addition to changes in transcript expression of genes directly related to TM function. These differences in conceptus development were supported by specific changes in composition of ULF in pregnant cows and by a larger increase in transcript expression of *RTP4* in PBL of pregnant multiparous cows in the OTM group. In addition to conceptus development, modest changes in the concentration of TM in follicular fluid and time to resume ovarian cyclicity, mostly in primiparous cows, were observed. Despite the effects on uterine and ovarian biology, no differences were observed in time to pregnancy, pregnancy risk, and late pregnancy losses, suggesting that the observed differences in preimplantation conceptus development might not be sufficient to improve pregnancy outcomes in lactating cows. Nonetheless, more studies evaluating the effects of TM supplementation strategies on reproductive biology and the performance of lactating cows in different management and oxidative stress conditions are warranted.

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ORCIDS

B. Mion https://orcid.org/0000-0002-7061-7415
G. Madureira https://orcid.org/0000-0002-4893-0966
J. F. W. Spricigo https://orcid.org/0000-0002-5507-3923
B. Van Winters https://orcid.org/0000-0002-9008-744X
S. J. LeBlanc https://orcid.org/0000-0003-2027-7704
M. A. Steele https://orcid.org/0000-0001-6941-6205
E. S. Ribeiro https://orcid.org/0000-0002-4201-7383